



Fig. 7: ESD tiles



Fig. 8: Bench-top ioniser



Fig. 9: Pink poly bag



Fig. 10: Black conductive poly bags



Fig. 11: Shield bag



Fig. 12: Moisture barrier bags



Fig. 13: Wrist strap and footwear tester

dissipative floors as long as the resistance to ground from anywhere on the floor is less than 1×10^9 ohms and personnel do not generate greater than 100V using the ANSI/ESD S97.2 test method.

There are various options available that you can choose

from, such as mats and tiles amongst others, but ensure that whatever you choose it complies with the standards mentioned above. Fig. 7 shows some ESD tiles.

ESD-protective tools and work station

When you are working with sensitive electronic components, you should consider buying all the equipment for your workstation that is marked as anti-static or ESD-safe. The materials mentioned below are optional but can be used for better EPA:

1. ESD table, chairs and stools
2. ESD-safe toolkit (cutter, plier and desoldering pump)
3. ESD-safe equipment such as soldering iron
4. ESD-safe brush
5. ESD-safe tray, bins and cabinets

A bench-top ioniser can also be used to neutralise electrostatic charges

at the workstation. This is the only ESD control method available to neutralise electrostatic charges on essential insulators or isolated conductors that may be at the workstation. The required limit according to ANSI/ESD S20.20 is less than $\pm 50V$ offset voltage (balance).

In addition to that, the discharge time to reduce +1000V to +100V and reduce -1000V to -100V should also be measured. Faster the static elimination time, the better. Do look for one that strictly complies with ANSI/ESD S20.20. Fig. 8 shows a bench-top ioniser.

Proper packaging. Place sensitive items in appropriate protective packaging when transporting or moving to storage. Static protective bags are really an integral part of a static control program. Selecting the appropriate bag can help reduce static damage and save money on costly repairs and rework. The cost of static protective packaging is insignificant as compared to the losses that may be there without them.

There are different types of bags available in the market, but just by looking at them you will not understand their capabilities. Here is how to distinguish them.

Pink poly bags (dissipative poly bags). These bags have the ability to dissipate a static charge to ground and keeps charge from building up on the package. The material is also anti-static, suggesting that it will not charge up by tribocharge. The material's resistivity

is in the dissipative range and is usually 10^9 to 10^{11} .

Unfortunately, these bags have no shielding ability. A static field or discharge occurring outside the bag will easily penetrate the bag and damage electronics inside.

Black conductive poly bags. Black poly bag is highly conductive as compared to other bags, usually about 10^3 to 10^6 , and will dissipate the charge very quickly. But, unfortunately, this fast dissipation also creates the possibility of spark at its surface. The material is conductive, therefore it does provide some small measure of shielding. However, there is no plastic layer (dielectric) to isolate a device inside a bag. The charge may be transferred through the volume of the material to the device instead of around the material to ground.

Shield bags. These provide the dissipative and anti-static attributes as well as add a metal shield and polyester dielectric to stop static from entering the bag. The test for shielding demonstrates the difference between the various bags. Shield bags will generally stop 97 per cent of a 1000V static pulse applied to the outside of the bag from reaching the inside. Pink poly bags will stop only about 10 per cent and black poly bags about 30 per cent.

Moisture barrier bags. These provide dissipation, anti-static and static shielding properties together with moisture vapour barrier. The moisture barrier protects moisture-sensitive items and improves long-term storage.

Then there are other products such as pink bubble bags, anti-static foam and anti-static boxes which are generally used in packaging for improving the reliability of the equipment in fields.

Wrist-strap/footwear tester

The wrist-strap/footwear tester is a highly recommended and widely used testing equipment that can be used to assure the compliance for proper grounding of the workers as required by ANSI/ESD S20.20. Consider buying one to complete your EPA. Fig. 13 shows a wrist-strap and footwear tester. ●

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